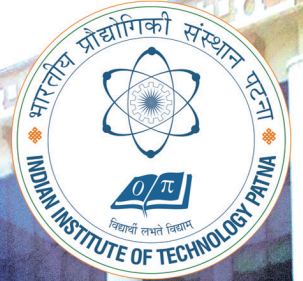


Indian Institute of Technology Patna



VLSI and Embedded Systems

Placement Brochure
2018 - 19

CURRICULUM

Core Courses

- ▶ Embedded Systems
- ▶ Digital VLSI Systems
- ▶ Analog & Mixed Signal Systems
- ▶ High Performance Computing Systems

Electives

- ▶ Cryptography
- ▶ Deep Learning
- ▶ VLSI Technology
- ▶ Sensors and Actuators
- ▶ Digital Image Processing
- ▶ Advanced Digital Signal Processing
- ▶ Radio Frequency Integrated Circuits
- ▶ Intelligent Visual Surveillance Systems
- ▶ Modeling & Simulation of MOS Devices

LABORATORIES

VLSI Laboratory

- | | |
|--------------------|-----------------|
| ⚙️ PADS | ⚙️ Vivado |
| ⚙️ Spectre | ⚙️ Xilinx ISE |
| ⚙️ Calibre | ⚙️ Cadence |
| ⚙️ Assura | ⚙️ Modelsim |
| ⚙️ Virtuoso | ⚙️ Visual TCAD |
| ⚙️ Mentor Graphics | ⚙️ Synopsys VCS |

Embedded System Laboratory

- | | |
|--------------------------|------------------|
| ⚙️ Arduino | ⚙️ Arduino IDE |
| ⚙️ RaspberryPi | ⚙️ MPLAB IDE |
| ⚙️ Nexys 4 DDR Artix-7 | ⚙️ ARM Cortex M3 |
| ⚙️ Xilinx Spartan-3E | ⚙️ IAR Workbench |
| ⚙️ PIC Development Board | |

Faculty



Dr. Kailash Chandra Ray

Research Area: VLSI & Embedded Systems



Dr. Pramod Kumar Tiwari

Research Area: Modeling & Simulation of Semiconductor Devices.



Dr. Jawar Singh

Research Area: Semiconductor Devices
Microelectronics, VLSI



Dr. Yatendra Kumar Singh

Research Area: RF , MEMS,
Computational Electromagnetics



Dr. Saurabh Kumar Pandey

Research Area: Optoelectronic Devices,
Semiconductor thin films

Dr. Arijit Mondal

Research Area: CAD for VLSI,
Analog EDA



Dr. Jimson Mathew

Research Area: Fault Tolerant
Computing, VLSI, Hardware Security



Dr. Somanath Tripathy

Research Area: Network Security



Dr. Maheshkumar H. Kolekar

Research Area: Image and Video
Processing, Video Surveillance



Dr. Rajib Kumar Jha

Research Area: Image and Video
Processing



Research Projects

1. Design and FPGA prototyping of multicarrier multiple access schemes for variable rate multimedia satellite communication.
2. Design and Implementation of Novel VLSI Architectures of PRNG for Cryptography Applications.
3. SMDP-C2SD.
4. Modeling, simulation and performance optimization of Re-S/D SOI MOSFET.
5. Analytical investigation of subthreshold behavior of SiNT FETs.
6. Exploration of 8/9 nano-meter process variation immune doping- and junction-free devices and their circuits.
7. Design and Development of RF Energy Harvesting Circuits for Low-power Electronic Devices.
8. Design and Analysis of High Performance RF MEMS-based Electronically Reconfigurable Filters for Wireless Communication Applications.

Internships



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Contact

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